



PIXOUS TECHNOLOGIES

Requirements Specification

HR Management Portal — Client Submission

Version 1.0 · 2026-08-14

Pixous HR Portal — Software Requirements Specification (SRS)

Field	Value
Project	Pixous HR Portal (My-EMP-Portal)
Version	1.0
Date	2026-08-14
Status	Approved for development (QA-audited)
Live deployment	http://16.192.105.61
Stack	Spring Boot 3 (Java 17) + React 19 + MySQL + Redis + Nginx

1. Introduction

1.1 Purpose

This document defines the complete functional and non-functional requirements of the Pixous HR Portal — a multi-tenant HR management system for IT services and field operations. It is the single source of truth for what the product must do, who may do it, and how it must behave.

1.2 Scope

The system covers the full employee lifecycle: onboarding, attendance, leave, payroll, assets, claims, complaints, helpdesk, safety, chat, reports, and administration — under a **5-role access model**.

1.3 Definitions & Abbreviations

Term	Meaning
Tenant / Company	An organisation isolated by <code>company_id</code> (e.g. "Pixous Technologies")
Role	A named set of authorities granted to a user
Authority	A granular permission (e.g. <code>USER_MANAGE</code>) checked by the backend
JWT	JSON Web Token used for stateless authentication
TL	Team Lead
HR	HR Manager
SA	System Admin (SUPER_ADMIN)

2. Roles & Access Control (5-Role Model)

2.1 Role Hierarchy

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SUPER_ADMIN (System Admin – platform-wide control)
├── COMPANY_ADMIN (Company Admin – one tenant)
│   ├── HR_MANAGER (HR – people operations)
│   │   └── TEAM_LEAD (TL – team operations)
│   └── EMPLOYEE (self-service)

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#	Role code	Display name	Description
R1	SUPER_ADMIN	System Admin	Platform owner: all tenants, all data, technical admin functions
R2	COMPANY_ADMIN	Company Admin	Administrator of a single company/tenant
R3	HR_MANAGER	HR	Employee lifecycle, leave & payroll operations
R4	TEAM_LEAD	Team Lead (TL)	Manages a team: approvals, team visibility
R5	EMPLOYEE	Employee	Self-service for own data and requests

2.2 Permission Matrix (Roles × Capabilities)

Capability	Authority	EMP	TL	HR	CA	SA
View own profile & update personal details	—	✓	✓	✓	✓	✓
Clock in / out (attendance)	—	✓	✓	✓	✓	✓
Apply for leave / view own balance	—	✓	✓	✓	✓	✓
View own payslips & download	PAYROLL_VIEW	✓	✓	✓	✓	✓
Submit claims & track status	—	✓	✓	✓	✓	✓
View & manage own assets	ASSET_MANAGE (self scope)	✓	✓	✓	✓	✓
Raise complaints / helpdesk tickets	—	✓	✓	✓	✓	✓
Chat & notifications	—	✓	✓	✓	✓	✓
Team dashboard, team attendance	ATTENDANCE_TEAM , DASHBOARD_EXEC	—	✓	✓	✓	✓
Approve team leave	LEAVE_APPROVE	—	✓	✓	✓	✓
Manage employees (CRUD)	EMPLOYEE_MANAGE	—	—	✓	✓	✓
Approve claims	CLAIM_APPROVE	—	—	✓	✓	✓
Run & approve payroll	PAYROLL_RUN , PAYROLL_APPROVE	—	—	✓	✓	✓
Manage leave policies	LEAVE_APPROVE (policy scope)	—	—	✓	✓	✓
View reports & export	REPORT_VIEW	—	—	✓	✓	✓
Manage users & roles	USER_MANAGE , ORG_MANAGE	—	—	—	✓	✓
Manage company config, branding, modules	ORG_MANAGE	—	—	—	✓	✓
Cross-tenant management, data reset	SUPER_ADMIN , TECHNICAL_ADMIN	—	—	—	—	✓
View audit logs (all)	AUDIT_VIEW	—	—	—	✓	✓

Rule: authorities are enforced **server-side** on every endpoint via `@PreAuthorize` ; the frontend only hides what the server rejects.

3. Functional Requirements

Each requirement has an ID, description, actor, and acceptance criteria (AC).

FR-01 Authentication & Session Management

ID	Requirement	Actor
FR-01.1	User signs in with username and password; system validates against stored hash	All
FR-01.2	On success, system issues an access JWT (short TTL) + refresh token (rotating)	All
FR-01.3	On failure, system records the attempt; 8 consecutive failures lock the account for that identity	All
FR-01.4	System rejects disabled accounts even with correct credentials	All
FR-01.5	User can change password; old password must be verified; new password ≥ 8 chars and different	All
FR-01.6	Password change revokes all other active sessions	All
FR-01.7	Access token expiry returns user to login; refresh flow restores session silently	All
FR-01.8	Logout revokes the refresh token	All

AC-01: logout must be per-identity and IP-spoof-proof; after logout the account stays locked until a successful login by an admin reset or a defined cooldown.

FR-02 Dashboard

ID	Requirement	Actor
FR-02.1	Employee dashboard shows personal summary: leave balance, attendance, payslips, pending approvals	EMP, TL, HR
FR-02.2	TL/HR dashboard shows team overview: headcount, attendance today, leave pending, birthdays/anniversaries	TL, HR, CA
FR-02.3	User count shown must be scoped to the active company , not platform totals	TL, HR, CA, SA
FR-02.4	Tech-admin dashboard shows platform totals and health	SA

FR-03 User & Employee Management

ID	Requirement	Actor
FR-03.1	Create employee: name, username, email, phone, department, designation, DOJ, DOB, reporting manager, role, employee code (auto), Aadhaar (validated)	HR, CA
FR-03.2	Edit employee details; change role; activate/deactivate account	HR, CA
FR-03.3	Search and filter employees (name, department, status) with pagination	HR, CA, TL (view)
FR-03.4	Import/export employees (XLSX) with validation report	HR, CA
FR-03.5	Prevent self-demotion and deletion of the last SUPER_ADMIN	SA
FR-03.6	Bulk assign employees to a company tenant	SA

FR-04 Leave Management

ID	Requirement	Actor
FR-04.1	Employee applies for leave: type, from-to dates, reason; validates balance and overlaps	EMP
FR-04.2	System computes leave balance per policy (e.g. annual, sick, casual)	All
FR-04.3	TL approves/rejects leave for direct reports; HR/CA can override	TL, HR, CA
FR-04.4	HR manages leave policies: types, accrual, max balance, restrictions	HR, CA
FR-04.5	Approval/ rejection sends notification to the applicant	All

FR-05 Attendance

ID	Requirement	Actor
FR-05.1	Employee clocks in and out; system records time and prevents duplicates	EMP
FR-05.2	Employee views own attendance calendar and monthly summary	EMP
FR-05.3	TL/HR view team attendance; HR can correct entries	TL, HR, CA
FR-05.4	System derives working hours from clock-in/out pairs	All

FR-06 Payroll & Payslips

ID	Requirement	Actor
FR-06.1	HR runs payroll for a month: selects employees, computes salary + deductions	HR, CA
FR-06.2	Payroll run is guarded against duplicate runs for the same period	HR, CA
FR-06.3	Generated payslips are visible to each employee (own only) and downloadable as PDF	EMP, HR, CA
FR-06.4	Payroll requests / approvals workflow for non-standard runs	HR, CA

FR-07 Assets

ID	Requirement	Actor
FR-07.1	Admin maintains asset catalogue and assigns assets to employees	HR, CA
FR-07.2	Employee views assets assigned to them and reports issues	EMP
FR-07.3	Asset lifecycle: issue, return, transfer; audit trail kept	HR, CA

FR-08 Claims & Reimbursement

ID	Requirement	Actor
FR-08.1	Employee submits a claim: type, amount, date, receipts (file upload)	EMP
FR-08.2	TL/HR approves or rejects; status notified	TL, HR
FR-08.3	Claim must not be editable after approval	All

FR-09 Complaints & Grievances

ID	Requirement	Actor
FR-09.1	Employee raises a complaint (anonymously if chosen) with category	EMP

FR-09.2	HR/CA tracks, responds, and closes complaints	HR, CA
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FR-10 Helpdesk & Support

ID	Requirement	Actor
FR-10.1	Employee opens a helpdesk ticket with priority	EMP
FR-10.2	Agent (HR/CA) claims, updates status, resolves with notes	HR, CA
FR-10.3	Ticket status changes notify the requester	All

FR-11 Safety

ID	Requirement	Actor
FR-11.1	Employees report safety incidents (type, location, description, attachments)	EMP
FR-11.2	HR/CA reviews and closes incidents; reports available	HR, CA

FR-12 Chat & Notifications

ID	Requirement	Actor
FR-12.1	Real-time chat (WebSocket/STOMP) between users; persisted history	All
FR-12.2	Notifications for approvals, mentions, and system events with unread count	All
FR-12.3	Notification preferences per module	All

FR-13 Calendar & Onboarding

ID	Requirement	Actor
FR-13.1	Shared calendar with events; employee sees own events	All
FR-13.2	Onboarding checklist: HR assigns tasks, employee completes them	HR, EMP

FR-14 Reports & Exports

ID	Requirement	Actor
FR-14.1	HR/CA generate reports: attendance summary, leave, payroll, headcount	HR, CA
FR-14.2	Reports exportable to XLSX (and payslips to PDF)	HR, CA
FR-14.3	Report access restricted by REPORT_VIEW	HR, CA, SA

FR-15 Audit Log

ID	Requirement	Actor
FR-15.1	System records auditable actions: actor, action, entity, timestamp	All
FR-15.2	CA/SA view and filter audit logs; logs are append-only	CA, SA

FR-16 Roles & Permissions

ID	Requirement	Actor
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FR-16.1	CA/SA create roles, assign authorities, and map roles to users	CA, SA
FR-16.2	Built-in roles (the 5 above) cannot be deleted	CA, SA
FR-16.3	Changes take effect on the user's next request (JWT authority refresh)	All

FR-17 Settings & Branding

ID	Requirement	Actor
FR-17.1	Company settings: name, industry, theme/branding (logo, colors)	CA
FR-17.2	Chatbot settings configurable	CA
FR-17.3	Profile settings: photo, contact details, password	All

FR-18 Multi-Tenancy & Companies

ID	Requirement	Actor
FR-18.1	Each company is isolated by <code>company_id</code> ; no cross-tenant data leakage	SA
FR-18.2	SA creates companies, assigns users, controls module enablement per company	SA
FR-18.3	New company bootstraps with the standard 5-role set	SA

FR-19 Data Reset (Technical Admin)

ID	Requirement	Actor
FR-19.1	SA can reset demo/tenant data with explicit confirmation	SA
FR-19.2	Reset is guarded by <code>SUPER_ADMIN</code> role and must never run against a live production DB automatically (seeding is prod-disabled)	SA

FR-20 File Uploads & Storage

ID	Requirement	Actor
FR-20.1	System accepts uploads (images, PDF, XLSX) with size limits	All
FR-20.2	Dangerous types (SVG, HTML, executables) are rejected or neutralised to <code>.bin</code>	All
FR-20.3	Filenames are never used in filesystem paths (traversal-proof)	All
FR-20.4	Downloads served with <code>nosniff</code> ; no directory listing	All

FR-21 AI Chatbot & Knowledge Base

ID	Requirement	Actor
FR-21.1	Employees can ask the assistant questions (leave policy, payroll, attendance) in the UI	All
FR-21.2	Answers are grounded in the company knowledge base and org context (<code>ChatbotKnowledge</code> , <code>ChatbotOrgContext</code>)	All
FR-21.3	Unanswerable queries return a graceful fallback message, never an exception	All

FR-22 Community, Calls & Presence

ID	Requirement	Actor
FR-22.1	Employees can post to the community feed and interact with posts	All
FR-22.2	Voice/video call initiation between employees with presence awareness	All
FR-22.3	Presence (online/away/offline) is broadcast in real time via event listeners and shown across the UI	All
FR-22.4	Chat events are produced/consumed asynchronously (Kafka-style event bus) without blocking the UI	All

FR-23 Performance Management

ID	Requirement	Actor
FR-23.1	HR/TL can define performance goals for employees (PerformanceGoal)	HR, TL
FR-23.2	HR can create performance reviews with ratings and comments (PerformanceReview)	HR
FR-23.3	Employees can view their goals and review history	EMP

FR-24 Tasks & Task Chat

ID	Requirement	Actor
FR-24.1	TL/HR can assign tasks to employees with status tracking (Task)	TL, HR
FR-24.2	Each task has a threaded chat for collaboration (TaskMessage)	All
FR-24.3	Task chat messages are persisted and real-time via the event bus	All

FR-25 Work Reports & Reminders

ID	Requirement	Actor
FR-25.1	Employees submit daily work reports with text and optional files	EMP
FR-25.2	TL/HR can review submitted reports (WorkReport)	TL, HR
FR-25.3	The reminder service nudges employees who have not submitted by the deadline	EMP

FR-26 Module Management & Cache (Technical Admin)

ID	Requirement	Actor
FR-26.1	SA can enable/disable application modules per company (MyModules , TechnicalAdminModule)	SA
FR-26.2	SA can clear application caches (Cache endpoints) for troubleshooting	SA
FR-26.3	Module toggles are persisted and respected by the UI and backend guards	SA

4. User Stories (by Role)

4.1 Employee (EMP)

US-EMP-01: As an employee, I can sign in and see my dashboard with my leave balance, today's attendance, and latest payslip.

US-EMP-02: As an employee, I can clock in/out and see my attendance history.

US-EMP-03: As an employee, I can apply for leave and track approval status.

- US-EMP-04: As an employee, I can submit a claim with receipt and track it.
- US-EMP-05: As an employee, I can raise complaints and helpdesk tickets.
- US-EMP-06: As an employee, I can chat with colleagues and get notifications.
- US-EMP-07: As an employee, I can change my password and update my profile.

4.2 Team Lead (TL)

- US-TL-01: As a TL, I can see my team's attendance and dashboard.
- US-TL-02: As a TL, I can approve/reject leave and claims from my direct reports.
- US-TL-03: As a TL, I can see my team's assigned assets.
- US-TL-04: As a TL, I cannot access payroll run, user management, or cross-team data.

4.3 HR Manager (HR)

- US-HR-01: As HR, I can manage the full employee lifecycle (create, edit, deactivate).
- US-HR-02: As HR, I can run and approve payroll and generate payslips.
- US-HR-03: As HR, I can manage leave policies and correct attendance.
- US-HR-04: As HR, I can generate and export reports.
- US-HR-05: As HR, I can manage assets, complaints, and safety incidents.

4.4 Company Admin (CA)

- US-CA-01: As Company Admin, I can manage users, roles, and permissions within my company.
- US-CA-02: As Company Admin, I can configure company settings and branding.
- US-CA-03: As Company Admin, I can view audit logs and all reports.
- US-CA-04: As Company Admin, I cannot access another company's data.

4.5 System Admin (SA)

- US-SA-01: As System Admin, I can create and manage companies (tenants).
- US-SA-02: As System Admin, I can access all tenants' data, audit logs, and technical functions (data reset, module management).
- US-SA-03: As System Admin, I can assign any role and manage platform-wide settings.

5. Non-Functional Requirements

NFR	Requirement	Target
NFR-01 Security	All endpoints enforce authn + authz server-side; JWT signed, short TTL, rotating refresh	No auth bypass; 401/403 correct
NFR-02 Security	Login rate limiting immune to header spoofing	Lockout at 8 failures
NFR-03 Security	Passwords hashed; no plaintext storage; no secrets in repo	—
NFR-04 Security	Upload allowlist + traversal protection	—
NFR-05 Performance	Page load (public) / login round-trip	< 1.5 s / < 1 s
NFR-06 Performance	List APIs paginated; p95	< 800 ms at 100 concurrent
NFR-07 Availability	Rolling 30-day uptime	≥ 99.5%
NFR-08 Usability	Responsive on desktop/tablet/mobile; dark + light mode	—

NFR-09 Maintainability	Layered architecture; automated tests gate releases	≥ 70% backend coverage
NFR-10 Compliance	Audit log of privileged actions; GDPR-friendly data access	—

6. Out of Scope (v1.0)

Public-facing candidate portal and job applications
Mobile native apps (PWA is provided)
Multi-currency payroll and tax filing integrations
BI/analytics beyond built-in reports
SSO/SAML integration (JWT-based auth only)

7. Requirements Traceability

7.1 Module → FR → Controllers

Module	FR IDs	Backend controllers	Primary roles
Auth & Sessions	FR-01	Auth, TechnicalAdminAuth	All
Dashboard	FR-02	Dashboard	All
Users & Employees	FR-03	User	HR, CA, SA
Leave	FR-04	Leave	All
Attendance	FR-05	Attendance	All
Payroll & Payslips	FR-06	Payroll	HR, CA
Assets	FR-07	Asset	All
Claims & Expenses	FR-08	TaExpense	All
Complaints	FR-09	Complaint	EMP, HR
Helpdesk	FR-10	Helpdesk	All
Safety	FR-11	SafetyIncident	All
Chat & Notifications	FR-12	Notification	All
Calendar & Onboarding	FR-13	Calendar, Onboarding	All
Reports & Exports	FR-14	Report	HR, CA
Audit Log	FR-15	Audit, TechnicalAdminAudit	CA, SA
Roles & Permissions	FR-16	Permission, TechnicalAdminRole	CA, SA
Settings & Branding	FR-17	Org, Settings	All
Multi-Tenancy	FR-18	TechnicalAdminCompany	SA
Data Reset	FR-19	DataReset	SA
Files & Storage	FR-20	File	All
AI Chatbot	FR-21	Chatbot	All

Community, Calls & Presence	FR-22	Community, Call, Presence	All
Performance	FR-23	Performance	HR, TL, EMP
Tasks & Task Chat	FR-24	Task	TL, HR, EMP
Work Reports	FR-25	WorkReport	EMP, TL, HR
Module Mgmt & Cache	FR-26	MyModules, TechnicalAdminModule, Cache	SA

7.2 Role → Modules

Role	Modules accessible
SUPER_ADMIN	All 26 modules incl. technical admin (FR-18, FR-19, FR-26) + all tenants
COMPANY_ADMIN	All business modules FR-01–FR-25 for own tenant + roles/permissions (FR-16) + audit (FR-15)
HR_MANAGER	FR-01–FR-14, FR-17, FR-20–FR-25 (people operations; no technical admin)
TEAM_LEAD	FR-01–FR-05, FR-07–FR-14, FR-20–FR-25 (team scope, approvals)
EMPLOYEE	FR-01–FR-05, FR-07–FR-14, FR-20–FR-25 (self-service only)

This document is version-controlled in the repository ([docs/REQUIREMENTS.md](#)). Any functional change must update this document and the corresponding unit tests.